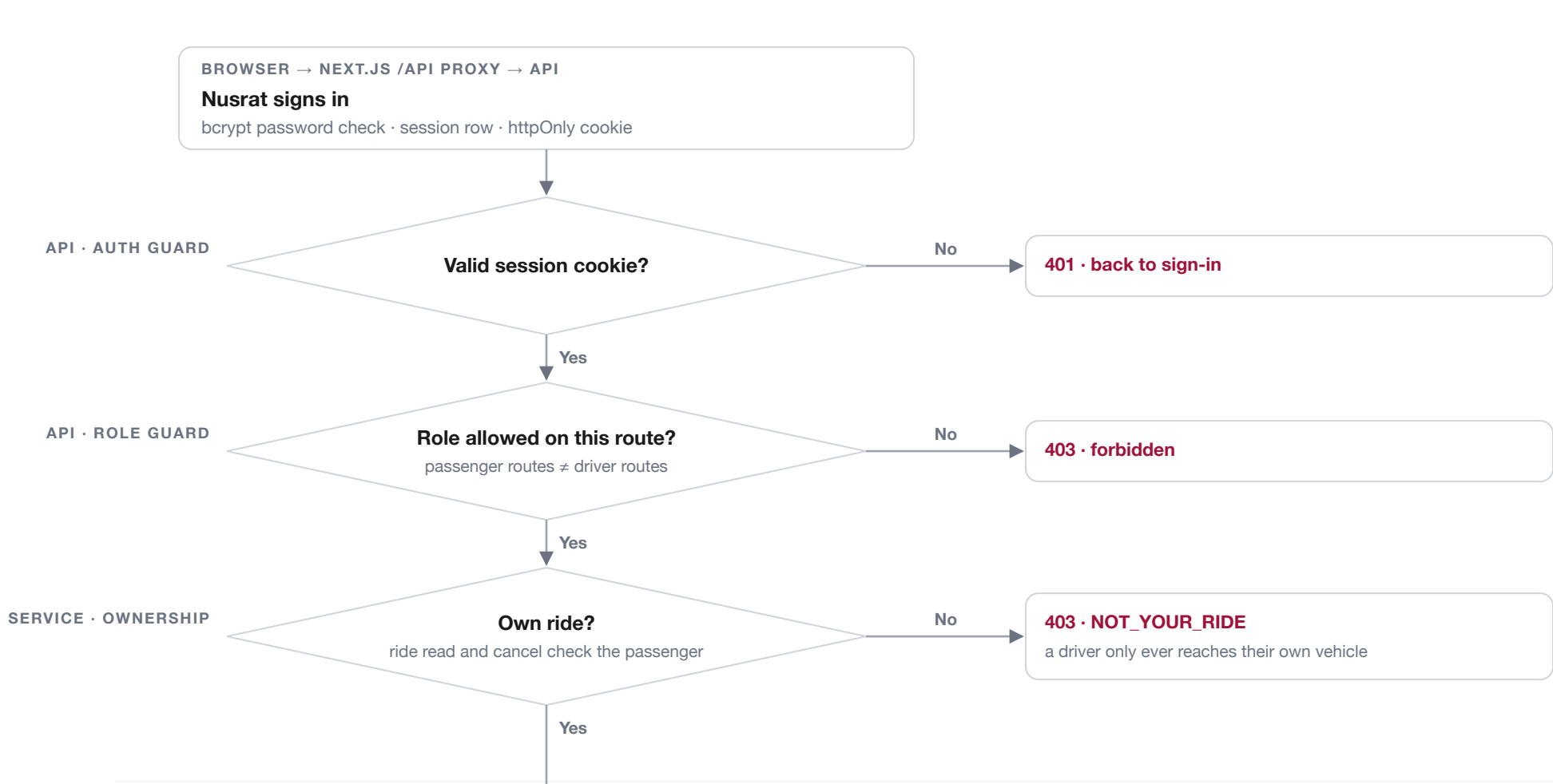


Dhaka Tesla Pool · Ride Flow

How one ride moves through the system · a Tesla picks people up along its route · side notes cover concurrent and edge cases

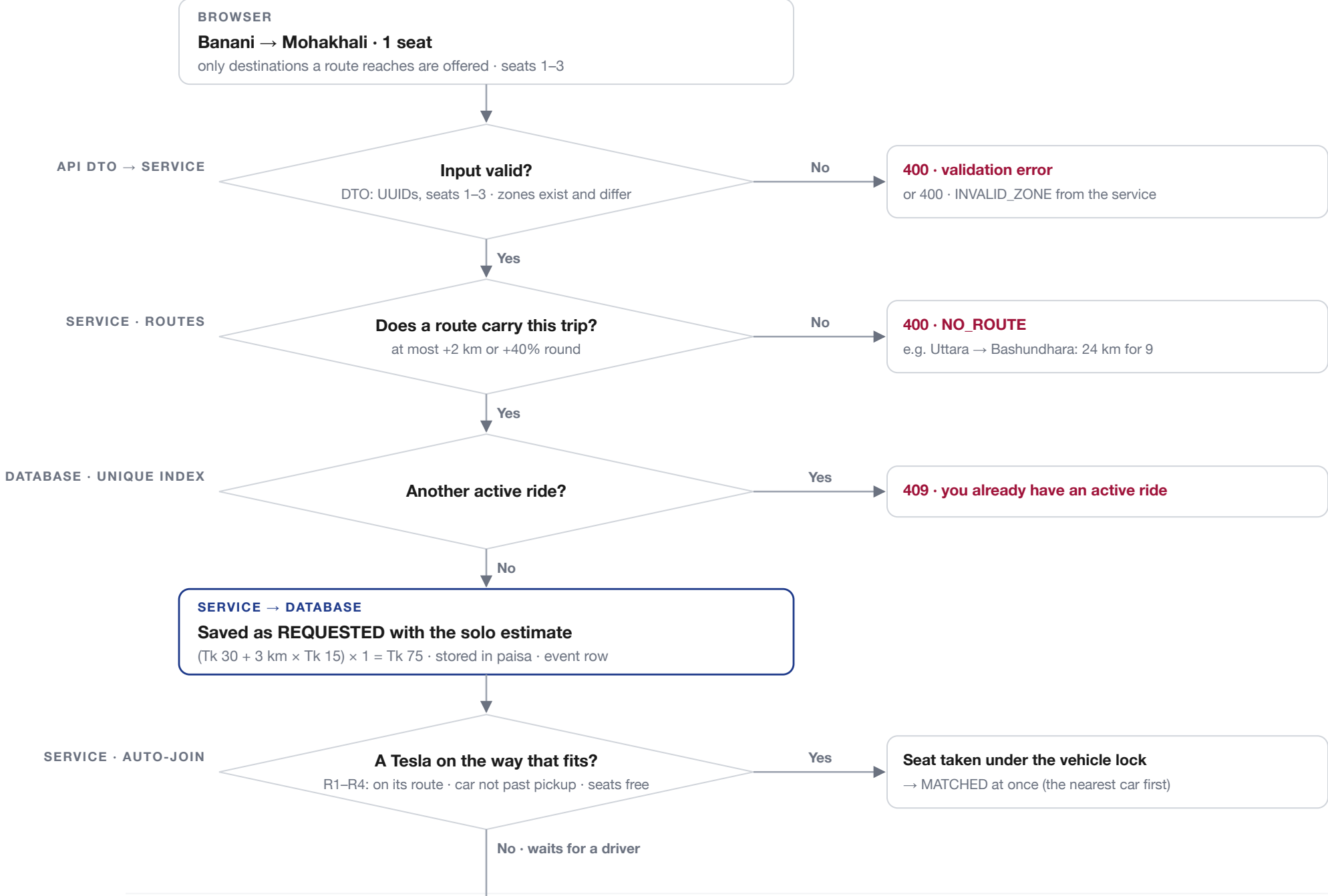
1 SIGN IN



Sign up

Passenger or driver, chosen first
Driver: NID or passport, licence, car
Unique email, phone, document, plate
bcrypt · 5 attempts per minute

2 REQUEST A RIDE



Routes (R1)

6 routes: 3 lines, both directions
At most +2 km or +40% round
Rafiq: 6 km for 4 km ✓
Uttara → Bashundhara: 24 for 9 ✗

Who pays, who earns

Rider: (Tk 30 + km × Tk 15) × seats
–20% if shared a hop · locked at drop-off
Driver: Tk 10/km carried + Tk 20/pickup
Platform keeps the rest (≥ Tk 10)
Late cancel Tk 20 → the driver who came

3 MATCHING



Last seat, many riders at once

Nusrat holds 2 of Bullet's 3 seats
20 riders ask for the last one
All wait for Bullet's lock
Exactly 1 wins · 19 keep waiting

No trip yet, two accepts at once

First accept starts the trip
Second waits, then joins it
One trip, two passengers

Retries and timeouts

Lock wait > 3 s → 503 BUSY
Auto-join is best effort: if busy,
the ride just keeps waiting

Database guarantees

seats_taken ≤ seat_capacity
pickup_stop < dropoff_stop
1 active pool per vehicle
1 active membership per ride
1 active ride per passenger
1 account per email, phone, ID, plate
Fee only on a cancelled ride

Route, location, online

Only routes through the car are listed
System suggests, driver confirms
Moving off the route clears it
Location follows the stops
Route changes only between trips
Offline refused while a pool is active

Leave vs join at the same stop

Car at Mohakhali, Shirin asks there
Both wait for Bullet's lock
Join first → the car waits for her
Leave first → stop is behind: she waits

What the passenger sees

Polls her ride every 3 s (history 10 s)
The route with the car on it
Own status and own fare only
Co-riders' first names only

Cancellation rules

Passenger: until picked up
Car coming to the stop: Tk 20
(2 min grace) · no-show too
→ paid with the next ride
Freed seat → next waiting rider
Twice → the same answer
Driver: before the first pickup →
passengers back to REQUESTED
Racing the driver: one lock, one wins

4 THE TRIP, STOP BY STOP

